

1 Search

Single trajectory t_0 ; read-only commands

ReAct loop: Thought $\rightarrow$ Action $\rightarrow$ Observation

LLM relevance: $r(o_i) = \text{LLM_score}(q, o_i) / 10$

Saturation: 3 consecutive $r(o_i) < 0.5$

Output: search report (files + context)



2 Strategy Proposal

LLM proposes $K = 5$ fix strategies $\{S_k\}$

Bidirectional NLI entailment clustering:

$$S_i \equiv S_j \Leftrightarrow P(e|S_i, S_j) > \theta \wedge P(e|S_j, S_i) > \theta$$

Semantic entropy: $H = -\sum_c p(c) \ln p(c)$

$H > \tau \Rightarrow$ fork one trajectory per cluster



3 Patch + Verify

Per-cluster trajectory in Docker container

SDLG: $\nabla_e \mathcal{L}_{\text{NLI}}$ attributes

high-impact tokens $\rightarrow$ substitute $\rightarrow N = 5$ alts

Cluster SDLG alternatives; branch if $H > \tau$

Verify: pytest $\rightarrow$ git diff $\rightarrow$ submit patch



4 Evaluation

SWE-bench harness per trajectory

FAIL_TO_PASS: bug-specific unit tests

PASS_TO_PASS: regression tests

$$\text{diverse-pass}@1 = \mathbb{1}[\exists t : \text{pass}(t)]$$

Hard cap: $B = 30$ trajectories per instance